higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING FABRICATION: BOILERMAKING
NQF LEVEL 4

22 NOVEMBER 2019

This marking guideline consists of 5 pages.

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SECTION A

QUESTION 1

1.1 A
1.2 D
1.3 B
1.4 C
1.5 B
1.6 A
1.7 B

(7 × 1) **[7]**

QUESTION 2

2.1 L
2.2 G
2.3 J
2.4 A
2.5 F
2.6 K
2.7 H
2.8 B
2.9 E
2.10 I

(10 × 1) **[10]**

QUESTION 3

3.1 Grinding
3.2 Bevel
3.3 Elbow
3.4 Reference point R

(4 × 1) **[4]**

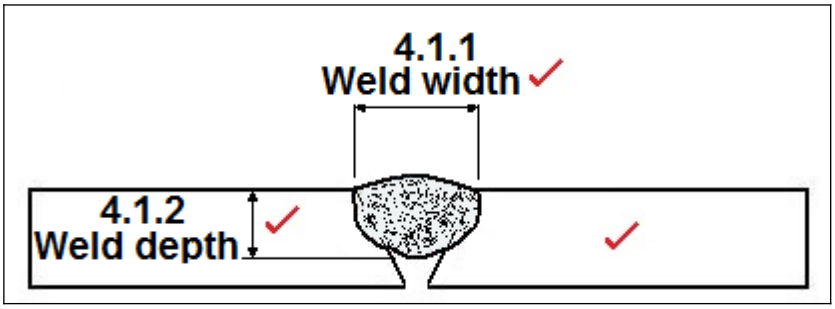
TOTAL SECTION A: 21

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SECTION B

QUESTION 4

4.1 4.1.1 and 4.1.2



(Any similar acceptable sketch)
(1 mark – weld width; 1 mark – weld depth; 1 mark – sketch) (3)

4.1.3 V-groove bevel area = $\frac{\text{bevel width} \times \text{bevel depth}}{2}$ ✓✓ (2)

4.2 Square head:
- To mark off 45° and 90° angles.
- To measure 45° and 90° angles.
- To use as spirit level.
- To use as a depth gauge. (Any 1 × 2)

Protractor head:
- To measure any angle of a slope.

Centre head:
- To find the centre of round objects.

(3 × 2) (6)

4.3
- Set the blade on the distance needed and lock the blade in position.
- Push the back mark gauge against the edge of the steel section.
- Use a scriber to scribe the center line of the hole, perpendicular to the edge. (3)

4.4 Mean diameter = OD – Mt
= 1 450 – 12 ✓
= 1 438 mm→ ✓

Length of plate = Dm × π
= 1438 × π ✓
= 4517,6 mm→✓ (4)

4.5
- Air
- Inert gas (2)

[20]

QUESTION 5

5.1
- General arrangement drawings
- Detail drawings
- Working drawings (3)

5.2 5.2.1 Isometric drawings indicate a three-dimensional drawing showing the front, side and top views in the same single view.

5.2.2 An orthographic projection is a two-dimensional drawing showing three views which indicate the height, depth and length of the object.

(2 × 2) (4)

5.3
- Chains for excessive wear or stretch
- Load and idler sprockets
- Hooks for cracks.
- Hook retaining nuts or collars, and pins and welds, or riveting used to secure the retaining members
- Brake mechanisms to see if they show glazed friction
- The rachets, pawls, and so on
- Bolts and nuts to see if they are loose
- Any parts, such as the hook block/s, chain attachments and
- bearings, to see if they are worn, cracked or distorted
- Supporting structures and trolleys to see if they can still support loads
- Electric and air hoists for excessive wear on the motor (6 × 2) (12)

[19]

QUESTION 6

6.1
- Instructions for the welder to fabricate the spool.
- The cut lengths of pipe.
- Any fittings and flanges needed to fabricate the spool.
- Materials required or specifications.
- Any special treatment needed.
- How many spools of the same type are required? (Any 3 × 1) (3)

6.2 The main difference between a pipefitter's square and a framing square are the added tables and the graduations on the pipefitter's square. (2)

6.3 Pressure = ρgh✓
= 1 100 × 9,8 × 60✓✓
= 646 800 Pa✓✓ OR 646,800 kPa✓
(Pa or kPa unit answers acceptable.) (5)

6.4 Given: Diameter = 120 mm
Therefore Radius = 60 mm✓
Formulae for Area of a circle = πr^2✓
Area of circle = $\pi \times 60^2$✓
$= \pi \times 3600$✓
$= 11310\ mm^2$✓ (5)

6.5
- All measuring and checking tools must be in good working order.
- Check the accuracy of the tools.
- Make sure the tools conform to the manufacturer's specifications. (3 × 2) (6)

[21]

QUESTION 7

7.1 It is a two-dimensional or three-dimensional path generated by the program code or CAM system. (2)

7.2
- Reference point R
- Tool reference point N
- Zero point of the machine M (3)

7.3
- Geometry
- Dimensions
- Tolerances
- Materials
- Surface finish
- Line types
- Hatching
- Projections (Any 7 × 1) (7)

7.4 Absolute movement:
- It is the movement to a position that was measured from a datum and all measurements are taken from that fixed point.

Incremental movement:
- It is the movement that is made from the previous position of the tool.

(2 × 2) (4)

7.5
- The type of material being cut.
- The speed and the feed that is used.
- The type of cutting tool that is used. (3)

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TOTAL SECTION B: 79
GRAND TOTAL: 100

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